Group-4

Software engineering

Unit testing

Backend Unit Testing:

Controllers:

- **AdminContoller:**

```
    getNotifications
      √ returns notifications (2 ms)
      √ skips status filter when empty string provided (2 ms)
      √ handles errors (28 ms)
    confirmNotification
      √ validates id (2 ms)
      √ requires existing notification (1 ms)
      √ prevents double processing (2 ms)
      √ updates payment and auction registrations (5 ms)
      √ updates payment even when type is not winning (2 ms)
      √ completes confirmation even when auction lookup fails (2 ms)
      √ handles primitive references and missing payment records (2 ms)
      √ skips registration when user already added (2 ms)
      √ handles confirmation failures (21 ms)
    rejectNotification
      √ validates id (6 ms)
      √ requires notification (1 ms)
      √ prevents double processing (2 ms)
      √ rejects payment and notification (1 ms)
      √ rejects notification without payment reference (3 ms)
      √ rejects primitive payment ids even if record missing (2 ms)
      √ handles rejection failures (31 ms)

Test Suites: 1 passed, 1 total
Tests:       44 passed, 44 total
Snapshots:   0 total
Time:        5.425 s
Ran all test suites matching tests/controllers/adminController.test.js.
```

- **AuctionController:**

```
PASS tests/controllers/auctionController.test.js
  auctionController
    uploadBase64Images
      √ returns 400 when no images provided (12 ms)
      √ saves images and responds with URLs (5 ms)
      √ reuses existing upload directory and falls back to http protocol (4 ms)
      √ handles unexpected failures (213 ms)
    createAuction
      √ creates auction and logs event (5 ms)
      √ normalizes images and metadata when payloads are invalid (3 ms)
      √ handles creation errors (97 ms)
    getMyAuctions
      √ returns paginated auctions (6 ms)
      √ handles errors (52 ms)
      √ applies status filters when provided (7 ms)
    getAuctionById
      √ rejects invalid id (7 ms)
      √ returns 404 when missing (5 ms)
      √ blocks unverified auctions for non-owners (4 ms)
      √ returns auction with top bids (6 ms)
      √ handles datastore errors (71 ms)
      √ allows owners to view unverified auctions when createdBy is primitive (3 ms)
    listAuctions
      √ lists verified auctions with filters (7 ms)
      √ normalizes condition filter strings (5 ms)
      √ handles errors while listing (60 ms)
    editAuction
      √ updates auction and logs event (7 ms)
      √ rejects invalid buy it now price (1 ms)
      √ returns 404 when auction missing on request (3 ms)
      √ recomputes status when schedule changes (16 ms)
      √ updates numeric increments when provided (4 ms)
      √ updates nested item fields even without title changes (3 ms)
      √ defaults invalid images payloads to an empty array (2 ms)
      √ reuses existing schedule boundaries when only one side is provided (12 ms)
```

```
    deleteAuction
      √ marks auction as cancelled (1 ms)
      √ returns 404 when auction not found (3 ms)
      √ handles delete errors (33 ms)
    handleRegisterInAuction
      √ rejects when auction missing (3 ms)
      √ rejects when user missing (2 ms)
      √ prevents seller from registering (2 ms)
      √ returns success and redirect URL (1 ms)
      √ handles unexpected registration errors (8 ms)

Test Suites: 1 passed, 1 total
Tests:       36 passed, 36 total
Snapshots:   0 total
Time:        4.349 s
Ran all test suites matching tests/controllers/auctionController.test.js.
```

- **AuthContoller:**

```
PASS tests/controllers/authController.test.js
  authController
    handleRegister
      √ validates required fields (11 ms)
      √ rejects duplicate email (2 ms)
      √ creates user and sends verification (3 ms)
      √ handles database errors (211 ms)
    handleLogin
      √ prevents double login (11 ms)
      √ requires email and password (5 ms)
      √ rejects missing user (3 ms)
      √ rejects invalid password (13 ms)
      √ requires verified user (4 ms)
      √ logs in successfully (4 ms)
      √ handles unexpected errors (81 ms)
    handleLogout
      √ clears token cookie (23 ms)
      √ handles logout failures (75 ms)
    verifyEmail
      √ validates input (2 ms)
      √ rejects invalid user (12 ms)
      √ verifies user and sends welcome email (11 ms)
      √ handles verification failures (47 ms)
    getCurrentUser
      √ returns null when no token user (2 ms)
      √ returns null when DB user missing (2 ms)
      √ returns sanitized user (1 ms)
      √ handles datasource failures (37 ms)
    handleResetPwdEmail
      √ requires email (3 ms)
      √ requires existing verified user (2 ms)
      √ sends reset email (7 ms)
      √ handles reset email failures (51 ms)
```

```
    handleResetPwd
      √ validates request body (5 ms)
      √ requires existing user (8 ms)
      √ validates token and matching passwords (3 ms)
      √ requires new password different from old (3 ms)
      √ resets password successfully (5 ms)
      √ handles reset failures (66 ms)
    getUserById
      √ requires an id parameter (2 ms)
      √ returns 404 when user is missing (3 ms)
      √ returns the public user payload (2 ms)
      √ handles database errors (68 ms)

Test Suites: 1 passed, 1 total
Tests:       35 passed, 35 total
Snapshots:   0 total
Time:        3.253 s
Ran all test suites matching tests/controllers/authController.test.js.
PS C:\Users\ADMIN\Desktop\Sofware Project\Repo5\BidSphere-Online-Auction-System\backend>
```

- **AutiBidController:**

```
PASS tests/controllers/autoBidController.test.js
  autoBidController
    setAutoBid
      √ rejects missing auction (14 ms)
      √ prevents duplicate auto bid (2 ms)
      √ creates auto bid entry (5 ms)
      √ avoids pushing duplicate auto bidders (3 ms)
      √ handles unexpected failures (91 ms)
    editAutoBid
      √ requires autobid id (3 ms)
      √ requires autobid record (2 ms)
      √ requires auction and active status (2 ms)
      √ updates max limit and logs event (12 ms)
      √ returns 500 when datastore fails (38 ms)
    activateAutoBid
      √ requires valid auction (6 ms)
      √ requires autobid (2 ms)
      √ returns early if already active (11 ms)
      √ rejects when max limit invalid (3 ms)
      √ sends outbid email if limit too low (3 ms)
      √ activates auto bid and triggers handler (12 ms)
      √ handles activation exceptions (39 ms)
    deactivateAutoBid
      √ requires autobid (2 ms)
      √ handles already inactive autobid (3 ms)
      √ deactivates autobid and updates auction (3 ms)
      √ handles deactivation failures (21 ms)

Test Suites: 1 passed, 1 total
Tests:       21 passed, 21 total
Snapshots:   0 total
Time:        2.686 s
Ran all test suites matching tests/controllers/autoBidController.test.js.
```

- **BidContoller:**

```
PASS tests/controllers/bidController.test.js
  bidController placeBid
    √ returns 404 when auction missing (11 ms)
    √ blocks manual bids when autobid active (2 ms)
    √ creates new bid and extends auction (4 ms)
    √ updates existing bid without extension (2 ms)
    √ handles unexpected errors (72 ms)

Test Suites: 1 passed, 1 total
Tests:       5 passed, 5 total
Snapshots:   0 total
Time:        2.706 s
Ran all test suites matching tests/controllers/bidController.test.js.
```

- **DeliveryContoller:**

```
PASS tests/controllers/deliveryController.test.js
  deliveryController
    createDelivery
      √ validates body (14 ms)
      √ requires auction and winner (3 ms)
      √ prevents duplicate delivery (2 ms)
      √ requires seller and valid payment (8 ms)
      √ uses default seller address and null payment id when paymentId missing (4 ms)
      √ validates payment ownership and auction match (2 ms)
      √ handles unexpected creation errors (108 ms)
      √ creates delivery record (4 ms)
      √ keeps payment status pending when payment capture has not succeeded (7 ms)
    getMyDeliveries
      √ returns deliveries for buyer or seller (4 ms)
      √ handles errors (182 ms)
    getAllDeliveries
      √ returns aggregated deliveries (5 ms)
      √ handles errors (51 ms)
    updateDeliveryStatus
      √ requires delivery (5 ms)
      √ requires seller authorization (29 ms)
      √ updates status fields (12 ms)
      √ saves even when optional fields are omitted (6 ms)
      √ handles errors (29 ms)
    updateUserAddress
      √ validates fields (4 ms)
      √ updates address (9 ms)
      √ handles address update failures (30 ms)

Test Suites: 1 passed, 1 total
Tests:       21 passed, 21 total
Snapshots:   0 total
```

- **PaymentContoller:**

```
PASS  tests/controllers/paymentController.test.js
  paymentController
    handleRegistrationPayment
      √ validates auction and user (15 ms)
      √ rejects when registration closed (3 ms)
      √ creates payment when live (10 ms)
      √ trims backend URL when generating verification link (3 ms)
      √ handles unexpected registration errors (4 ms)
    handleWinningCodPayment
      √ requires auction, user, and winner (6 ms)
      √ creates COD payment and notification (4 ms)
      √ handles COD flow failures (2 ms)
    handleWinningUpiPayment
      √ validates inputs similarly to COD (4 ms)
      √ creates UPI payment (10 ms)
      √ handles UPI flow failures (16 ms)
    verifyPayment
      √ validates user and payment (3 ms)
      √ skips already verified payments (8 ms)
      √ submits verification request (2 ms)
      √ handles verification errors (3 ms)
    getPaymentById
      √ handles missing payment (3 ms)
      √ returns payment data (2 ms)
      √ handles payment lookup errors (2 ms)
    getMyPayments
      √ requires authentication (2 ms)
      √ returns payments (1 ms)
      √ handles errors (249 ms)

Test Suites: 1 passed, 1 total
Tests:       21 passed, 21 total
Snapshots:   0 total
```

- **RatingController:**

```
PASS  tests/controllers/ratingController.test.js
  ratingController
    rateSeller
      √ validates required fields (12 ms)
      √ validates rating range (4 ms)
      √ returns 404 when auction missing (3 ms)
      √ rejects ratings for non-ended auctions (2 ms)
      √ requires the auction winner to rate (4 ms)
      √ prevents rating yourself (2 ms)
      √ prevents duplicate rating (4 ms)
      √ creates rating and updates seller stats (9 ms)
      √ defaults to zeroed stats when aggregate empty (9 ms)
      √ handles errors (158 ms)
    getSellerRatings
      √ returns ratings list (9 ms)
      √ supports non-objectId seller filters (2 ms)
      √ handles errors (119 ms)
    updateRating
      √ validates rating id (4 ms)
      √ validates rating range (12 ms)
      √ returns 404 when rating missing (7 ms)
      √ enforces owner (1 ms)
      √ updates rating and seller stats (2 ms)
      √ updates rating alone when no review is provided (10 ms)
      √ updates review only and handles empty aggregates (3 ms)
      √ retains review updates even when clearing the text (6 ms)
      √ handles update errors (53 ms)
```

```
    deleteRating
      √ validates rating id (3 ms)
      √ returns 404 for missing rating (2 ms)
      √ enforces owner on delete (2 ms)
      √ deletes rating and refreshes seller stats (3 ms)
      √ recomputes seller stats with aggregate data (5 ms)
      √ handles delete errors (57 ms)

Test Suites: 1 passed, 1 total
Tests:       28 passed, 28 total
Snapshots:   0 total
Time:        4.276 s
Ran all test suites matching tests/controllers/ratingController.test.js.
```

- **UserController:**

```
PASS tests/controllers/userController.test.js
  userController
    getBiddingHistory
      √ requires authentication (14 ms)
      √ returns aggregated history (4 ms)
      √ falls back to zero total and default auction fields (4 ms)
      √ handles errors (199 ms)
    getWatchlist
      √ requires authentication (5 ms)
      √ returns watchlist items (4 ms)
      √ handles entries missing populated auction data (2 ms)
      √ falls back to top-level auction images when item missing (10 ms)
      √ defaults images to empty array when auction data lacks images (18 ms)
      √ handles errors (93 ms)
    addToWatchlist
      √ validates auth and payload (6 ms)
      √ requires auction existence (6 ms)
      √ returns existing watchlist id (20 ms)
      √ creates new watchlist entry (4 ms)
      √ handles creation failures (52 ms)
    removeFromWatchlist
      √ validates auth and params (2 ms)
      √ deletes watchlist entry (8 ms)
      √ handles deletion failures (36 ms)
```

```
    updateUserProfile
      √ requires authentication (2 ms)
      √ returns 404 when user missing (2 ms)
      √ updates profile with optional fields (2 ms)
      √ handles update errors (28 ms)

Test Suites: 1 passed, 1 total
Tests:       22 passed, 22 total
Snapshots:   0 total
Time:        4.704 s
Ran all test suites matching tests/controllers/userController.test.js.
```

- **Coverage_report:**

All files

100% Statements 881/881 **100%** Branches 561/561 **100%** Functions 58/58 **100%** Lines 833/833

Press *n* or *j* to go to the next uncovered block, *b*, *p* or *k* for the previous block.

Filter:

File	Statements		Branches		Functions		Lines	
adminController.js	100%	150/150	100%	90/90	100%	11/11	100%	141/141
auctionController.js	100%	170/170	100%	125/125	100%	10/10	100%	154/154
authController.js	100%	116/116	100%	67/67	100%	8/8	100%	114/114
autoBidController.js	100%	89/89	100%	35/35	100%	5/5	100%	88/88
bidController.js	100%	32/32	100%	13/13	100%	1/1	100%	32/32
deliveryController.js	100%	76/76	100%	59/59	100%	5/5	100%	73/73
paymentController.js	100%	99/99	100%	37/37	100%	7/7	100%	97/97
ratingController.js	100%	75/75	100%	57/57	100%	4/4	100%	73/73
userController.js	100%	74/74	100%	78/78	100%	7/7	100%	61/61

Jobs:

- **PaymentStatusJob:**

```
PS C:\Users\kakad\OneDrive\Desktop\Payment\BidSphere-Online-Auction-System\backend> npx
 jest c:\Users\kakad\OneDrive\Desktop\Payment\BidSphere-Online-Auction-System\backend\t
est_for_jobs\paymentStatusJob.test.js
 PASS  test_for_jobs/paymentStatusJob.test.js
  startPaymentStatusJob
    √ throws on invalid cron pattern (33 ms)
    √ schedules cron with defaults (3 ms)
    √ prevents duplicate scheduling (2 ms)
    √ runs immediate sync when defaults are used (3 ms)
    √ skips initial sync when runOnStart false (4 ms)
  stopPaymentStatusJob
    √ stops active cron job (1 ms)
    √ is no-op when job not running (1 ms)
  updateExpiredPayments via scheduler
    √ handles case with no expired payments (2 ms)
    √ marks expired payments as failed (2 ms)
    √ processes multiple batches (2 ms)
    √ resets processing flag after exceptions (2 ms)
    √ prevents overlapping runs (2 ms)
    √ reports processing state while running (2 ms)
    √ is resilient when updateMany throws (2 ms)
    √ handles scenario with pending payments lacking ids (2 ms)
    √ keeps running flag true while cron scheduled (1 ms)

Test Suites: 1 passed, 1 total
Tests:       16 passed, 16 total
Snapshots:   0 total
Time:        0.681 s, estimated 1 s
Ran all test suites matching c:\Users\kakad\OneDrive\Desktop\Payment\BidSphere-Online-A
uction-System\backend\test_for_jobs\paymentStatusJob.test.js.
```

- **RegistrationStatusJob:**

```
PASS test_for_jobs/registrationStatusJob.test.js
  startRegistrationStatusJob
    √ throws when cron pattern invalid (36 ms)
    √ schedules a job with provided cron pattern (23 ms)
    √ does not schedule twice if already running (5 ms)
    √ runs the initial sync when defaults are used (7 ms)
    √ skips the initial sync when runOnStart is false (5 ms)
  stopRegistrationStatusJob
    √ stops existing cron job and clears status (5 ms)
    √ is a no-op when no cron job is running (2 ms)
  updateRegistrationWindows via scheduler
    √ exits early when already processing (4 ms)
    √ processes batches and skips when list empty (3 ms)
    √ updates auctions when registration flag differs (6 ms)
    √ does not send updates when flags already correct (4 ms)
    √ handles multiple batches when batch size reached (7 ms)
    √ logs and continues when bulkWrite throws (5 ms)
    √ resets processing flag after errors (3 ms)
    √ reports processing state while handler is running (6 ms)
    √ prevents overlapping runs (5 ms)
    √ handles auctions closing registration window (6 ms)
    √ propagates console.error when updateRegistrationWindows throws (5 ms)

Test Suites: 1 passed, 1 total
Tests:       18 passed, 18 total
Snapshots:   0 total
Time:        0.763 s, estimated 1 s
Ran all test suites matching c:\Users\kakad\OneDrive\Desktop\Payment\BidSphere-Online-A
uction-System\backend\test_for_jobs\registrationStatusJob.test.js.
```

- **AuctionStatusUpadater:**

```
PS C:\Users\kakad\OneDrive\Desktop\Payment\BidSphere-Online-Auction-System\backend> npx
 jest c:\Users\kakad\OneDrive\Desktop\Payment\BidSphere-Online-Auction-System\backend\t
est_for_jobs\auctionStatusUpdater.test.js
 PASS  test_for_jobs/auctionStatusUpdater.test.js
  startAuctionStatusUpdater
    √ rejects invalid cron pattern (47 ms)
    √ schedules job with cron pattern (3 ms)
    √ does not start twice (2 ms)
    √ runs immediate sync when defaults are used (3 ms)
    √ skips immediate sync when runOnStart false (11 ms)
    √ logs initial run errors via the catch handler (5 ms)
    √ logs scheduled run errors via the catch handler (3 ms)
  stopAuctionStatusUpdater
    √ calls stop on cron job (2 ms)
    √ does nothing when not running (2 ms)
  updateAuctionStatuses via scheduler
    √ handles no auctions gracefully (3 ms)
    √ updates UPCOMING auctions to LIVE (3 ms)
    √ updates UPCOMING auctions directly to ENDED when endTime passed (1 ms)
    √ transitions LIVE auctions to ENDED (2 ms)
    √ sets winners based on highest bids (2 ms)
    √ skips winner update when no bids (2 ms)
    √ fills winners for ENDED auctions missing winner info (2 ms)
    √ leaves ENDED auctions without bids untouched in the final sweep (2 ms)
    √ breaks batch loops when batch smaller than size (8 ms)
    √ logs errors and resets isProcessing flag (3 ms)
    √ prevents overlapping execution (1 ms)
    √ reports processing status while running (2 ms)
    √ logs summary when updates occur (5 ms)
    √ skips bulk logging when batch IDs collapse to empty (1 ms)
    √ skips winner fixes when all ended auctions already have winners (1 ms)

Test Suites: 1 passed, 1 total
Tests:       24 passed, 24 total
Snapshots:   0 total
Time:        0.748 s, estimated 2 s
Ran all test suites matching c:\Users\kakad\OneDrive\Desktop\Payment\BidSphere-Online-A
uction-System\backend\test_for_jobs\auctionStatusUpdater.test.js.
```

- # Coverage_report

All files jobs

100% Statements 162/162 **100%** Branches 79/79 **100%** Functions 22/22 **100%** Lines 146/146

Press *n* or *j* to go to the next uncovered block, *b*, *p* or *k* for the previous block.

Filter:

File		Statements		Branches		Functions		Lines	
au-registrationStatusJob.js		100%	49/49	100%	25/25	100%	5/5	100%	45/45
auctionStatusUpdater.js		100%	72/72	100%	35/35	100%	11/11	100%	65/65
paymentStatusJob.js		100%	41/41	100%	19/19	100%	6/6	100%	36/36

Code coverage generated by istanbul at 2025-11-22T19:58:12.478Z

```
PASS  test_for_jobs/auctionStatusUpdater.test.js
--------------------------|---------|----------|---------|---------|-------------------
File                      | % Stmts | % Branch | % Funcs | % Lines | Uncovered Line #s
--------------------------|---------|----------|---------|---------|-------------------
All files                 |     100 |      100 |     100 |     100 |
 ...istrationStatusJob.js |     100 |      100 |     100 |     100 |
 auctionStatusUpdater.js  |     100 |      100 |     100 |     100 |
 paymentStatusJob.js      |     100 |      100 |     100 |     100 |
--------------------------|---------|----------|---------|---------|-------------------

Test Suites: 3 passed, 3 total
Tests:       58 passed, 58 total
Snapshots:   0 total
Time:        1.513 s, estimated 3 s
Ran all test suites matching c:\Users\kakad\OneDrive\Desktop\Payment\BidSphere-Online-A
uction-System\backend\test_for_jobs.
```

Middleware:

- **BidValidMiddleware:**

```
PASS  tests/middleware/bidValidMiddleware.test.js
  validateBid
    √ returns 404 when the auction is missing (16 ms)
    √ requires numeric amounts (3 ms)
    √ requires positive amounts (3 ms)
    √ requires bids above current and starting price (4 ms)
    √ requires bids to respect the min increment (3 ms)
    √ prevents sellers from bidding on their auctions (6 ms)
    √ calls next for valid bids (2 ms)
    √ returns 500 when the lookup fails (100 ms)
  validateAutoBid
    √ requires numeric limits (12 ms)
    √ returns 400 when the auction is missing (3 ms)
    √ prevents sellers from creating auto-bids (3 ms)
    √ ensures the max limit beats the current bid + increment (2 ms)
    √ calls next when validations pass (2 ms)
    √ returns 500 on failures (31 ms)

Test Suites: 1 passed, 1 total
Tests:       14 passed, 14 total
Snapshots:   0 total
Time:        3.875 s
Ran all test suites within paths "tests/middleware/bidValidMiddleware.test.js".
```

- **AuthMiddleware:**

```
PASS  tests/middleware/authMiddleware.test.js
  authMiddleware
    restrictToLoggedinUserOnly
      √ requires the auth cookie (11 ms)
      √ rejects when token user is missing (3 ms)
      √ attaches the user and calls next (3 ms)
      √ returns 500 when getUser throws (83 ms)
    checkAuth
      √ silently loads the user when present (3 ms)
      √ continues when no token exists (5 ms)
      √ returns 500 on errors (12 ms)

Test Suites: 1 passed, 1 total
Tests:       7 passed, 7 total
Snapshots:   0 total
Time:        2.406 s
Ran all test suites within paths "tests/middleware/authMiddleware.test.js".
```

- **AuctionValidMiddleware**

```
  ensureBeforeStart
    √ requires the auction to be loaded (2 ms)
    √ rejects once the auction has started (9 ms)
    √ enforces minimum days before start (11 ms)
    √ allows operations before the deadline (20 ms)
    √ allows operations when no buffer is required (2 ms)

Test Suites: 1 passed, 1 total
Tests:       29 passed, 29 total
Snapshots:   0 total
Time:        7.291 s
Ran all test suites within paths "tests/middleware/auctionValidMiddleware.test.js".
```

```
PASS tests/middleware/auctionValidMiddleware.test.js (6.455 s)
  validateCreateAuction
    √ passes with a valid payload (67 ms)
    √ allows auctions without a buy-it-now price (9 ms)
    √ requires a title (13 ms)
    √ requires an item name (2 ms)
    √ validates starting price (11 ms)
    √ validates min increment (3 ms)
    √ requires start and end times (3 ms)
    √ rejects invalid dates (13 ms)
    √ rejects when end time is before start time (4 ms)
    √ validates buyItNowPrice positivity (4 ms)
    √ requires buyItNowPrice to exceed starting price (3 ms)
  validateUpdateAuction
    √ rejects unknown fields (5 ms)
    √ rejects invalid start/end combinations (8 ms)
    √ validates single date fields (6 ms)
    √ validates single endTime values (2 ms)
    √ validates numeric fields (4 ms)
    √ requires images to be arrays (6 ms)
    √ validates condition values (2 ms)
    √ accepts whitelisted condition values (2 ms)
    √ allows valid buyItNowPrice updates (1 ms)
    √ accepts updates when start and end times are valid (1 ms)
    √ passes valid updates (1 ms)
  validateObjectIdParam
    √ rejects invalid ids (3 ms)
    √ passes valid ids (1 ms)
```

- **AdminMiddleware:**

```
PASS tests/middleware/adminMiddleware.test.js
  adminMiddleware
    restrictAdminIP
      √ allows requests from the configured IP (99 ms)
      √ normalizes IPv4-mapped IPv6 addresses (15 ms)
      √ rejects traffic from other IPs (19 ms)
      √ handles unexpected errors when IP is missing (27 ms)
    requireAdminAuth
      √ allows requests with a valid admin token (3 ms)
      √ rejects missing tokens (3 ms)
      √ rejects invalid tokens (6 ms)
      √ returns 500 when cookie access throws (15 ms)

Test Suites: 1 passed, 1 total
Tests:       8 passed, 8 total
Snapshots:   0 total
Time:        2.929 s
Ran all test suites within paths "tests/middleware/adminMiddleware.test.js".
```

- **TokenValidMiddleware:**

```
PASS tests/middleware/tokenValidMiddleware.test.js
  validateRegistration
    √ returns 404 when the auction is missing (14 ms)
    √ rejects when the user has not registered (6 ms)
    √ calls next when the user is registered (3 ms)
    √ returns 500 when database lookup throws (130 ms)

Test Suites: 1 passed, 1 total
Tests:       4 passed, 4 total
Snapshots:   0 total
Time:        2.342 s
Ran all test suites within paths "tests/middleware/tokenValidMiddleware.test.js".
```

- **OwnMiddleware:**

```
PASS tests/middleware/ownMiddleware.test.js
  validateAuctionOwnership
    √ requires an auction id parameter (15 ms)
    √ requires authentication (3 ms)
    √ returns 404 when the auction is missing (6 ms)
    √ rejects when the user is not the owner (4 ms)
    √ attaches the auction when the user owns it (2 ms)
    √ returns 400 when the database call throws (256 ms)

Test Suites: 1 passed, 1 total
Tests:       6 passed, 6 total
Snapshots:   0 total
Time:        4.885 s
Ran all test suites within paths "tests/middleware/ownMiddleware.test.js".
```

- # CoverageReport of Middleare:

All files **middleware**

100% Statements 172/172 **100%** Branches 141/141 **100%** Functions 16/16 **100%** Lines 171/171

Press *n* or *j* to go to the next uncovered block, *b*, *p* or *k* for the previous block.

Filter:

File		Statements		Branches		Functions		Lines	
adminMiddleware.js		100%	19/19	100%	8/8	100%	2/2	100%	19/19
auctionValidMiddleware.js		100%	70/70	100%	91/91	100%	7/7	100%	69/69
authMiddleware.js		100%	18/18	100%	4/4	100%	2/2	100%	18/18
bidValidMiddleware.js		100%	36/36	100%	26/26	100%	2/2	100%	36/36
ownMiddleware.js		100%	16/16	100%	8/8	100%	1/1	100%	16/16
tokenValidMiddleware.js		100%	13/13	100%	4/4	100%	2/2	100%	13/13

Services:

- **Auth:**

```
    √ returns user object when found (1 ms)
    √ returns null if user not found (2 ms)
    √ propagates error if User.findById throws BEFORE select is accessed (4 ms)
    √ propagates error if .select() rejects (2 ms)
    √ handles case where jwt.verify returns payload with extra fields (2 ms)
    √ works when payload._id is numeric (1 ms)
    √ can be called multiple times and reuses mocks (1 ms)
    √ does not modify the token string passed in (1 ms)
    √ jwt.verify is only called once per getUser call (1 ms)
    √ does not call User.findById when token invalid (1 ms)
    √ handles token that yields payload with only _id (1 ms)
  generateHashPassword
    √ calls bcrypt.hash with the password and salt rounds 10 (2 ms)
    √ returns the hash value from bcrypt.hash (1 ms)
    √ can hash different passwords independently (1 ms)
    √ propagates error when bcrypt.hash rejects (2 ms)
    √ allows empty string password (1 ms)
    √ allows very long password (1 ms)

Test Suites: 1 passed, 1 total
Tests:       42 passed, 42 total
Snapshots:   0 total
Time:        1.807 s
```

```
PS C:\Users\kakad\OneDrive\Desktop\Payment\BidSphere-Online-Auction-System\backend> npx j
est c:\Users\kakad\OneDrive\Desktop\Payment\BidSphere-Online-Auction-System\backend\test_
for_services\auth.test.js
 PASS  test_for_services/auth.test.js
  setUser
    √ signs a JWT with _id and email (8 ms)
    √ returns whatever jwt.sign returns (1 ms)
    √ uses secret '$@BidSphere' exactly (1 ms)
    √ ignores extra fields on user when building token payload (2 ms)
    √ works if email is null (still passes email: null) (2 ms)
    √ works if email is undefined (still passes email: undefined) (3 ms)
    √ throws if jwt.sign throws (66 ms)
    √ supports numeric-like string id (1 ms)
    √ supports non-string id (e.g., ObjectId-like) (1 ms)
    √ allows empty string email (1 ms)
    √ allows empty string _id (2 ms)
    √ still works even if user object has no _id and no email (2 ms)
  getUser
    √ returns null if token is null (3 ms)
    √ returns null if token is undefined (1 ms)
    √ returns null if token is empty string (2 ms)
    √ returns null if jwt.verify throws generic error (1 ms)
    √ returns null if jwt.verify throws TokenExpiredError-like (1 ms)
    √ returns null if payload is null (2 ms)
    √ returns null if payload is not an object (2 ms)
    √ returns null if payload._id is missing (1 ms)
    √ returns null if payload._id is null (1 ms)
    √ returns null if payload._id is empty string (2 ms)
    √ calls jwt.verify with secret '$@BidSphere' (2 ms)
    √ calls User.findById with payload._id (1 ms)
    √ uses correct projection when selecting user fields (1 ms)
    √ returns user object when found (1 ms)
    √ returns null if user not found (2 ms)
    √ propagates error if User.findById throws BEFORE select is accessed (4 ms)
    √ propagates error if .select() rejects (2 ms)
    √ handles case where jwt.verify returns payload with extra fields (2 ms)
    √ works when payload._id is numeric (1 ms)
    √ can be called multiple times and reuses mocks (1 ms)
```

- **AutoBid:**

```
PS C:\Users\kakad\OneDrive\Desktop\Payment\BidSphere-Online-Auction-System\backend> npx jest c:\Users\kakad\OneDrive\Desktop\Payment\B
idSphere-Online-Auction-System\backend\test_for_services\autoBid.test.js
 PASS  test_for_services/autoBid.test.js
  handleAutoBids
    √ returns early if auction not found or not LIVE (11 ms)
    √ returns early if there are no autobidders (5 ms)
    √ skips bidder who is already the current winner (4 ms)
    √ sends outbid email and does not place bid when nextBid > maxLimit (3 ms)
    √ creates new bid, updates auction & logs AUTO_BID_TRIGGERED when within maxLimit and no existing bid (5 ms)
    √ extends auction endTime and logs AUCTION_EXTENDED when within last 5 minutes (4 ms)

Test Suites: 1 passed, 1 total
Tests:       6 passed, 6 total
Snapshots:   0 total
Time:        2.281 s
Ran all test suites matching c:\Users\kakad\OneDrive\Desktop\Payment\BidSphere-Online-Auction-System\backend\test_for_services\autoBid
.test.js.
```

- **Cloudinary:**

```
PASS test_for_services/cloudinary.service.test.js
  cloudinary.service
    √ calls cloudinary.config with environment values on import (37 ms)
    √ exposes sanitizeFilename default branch via __testables (9 ms)
    √ exposes buildPublicId default branch via __testables (8 ms)
    √ uploadBase64ToCloudinary - success returns expected shape and forwards options (11
ms)
    √ uploadBase64ToCloudinary - propagates upload errors as thrown Error (117 ms)
    √ uploadBase64ToCloudinary - fallback error message used when cloudinary error lacks
message (16 ms)
    √ uploadBase64ToCloudinary - uses default folder when not provided (11 ms)
    √ uploadBase64ToCloudinary - when no filename provided uses default 'file' in public_
id (14 ms)
    √ uploadBufferToCloudinary - success via upload_stream and streamifier piping (9 ms)
    √ uploadBufferToCloudinary - upload_stream invokes callback with error -> rejects (39
 ms)
    √ uploadBufferToCloudinary - upload_stream error fallback message when callback error
 lacks message (17 ms)
    √ uploadBufferToCloudinary - streamifier.createReadStream throws -> rejects (25 ms)
    √ deleteFromCloudinary - calls destroy with invalidate true when publicId provided (9
 ms)
    √ deleteFromCloudinary - returns early when publicId falsy (7 ms)
    √ deleteFromCloudinary - propagates destroy error (19 ms)
    √ deleteFromCloudinary - fallback error message used when destroy error lacks message
 (13 ms)
    √ buildPublicId removes extension and sanitizes filename via options.public_id (9 ms)
    √ uploadBase64ToCloudinary - maps all returned fields (bytes,width,height,format) (7
ms)
    √ module import warns when credentials missing (13 ms)
    √ module import uses default CLOUDINARY_FOLDER when env missing (12 ms)
    √ module import warns when individual credentials missing (name, key, secret) (15 ms)
    √ uploadBase64ToCloudinary - filename without extension is handled (3 ms)
    √ uploadBufferToCloudinary - uploader throws synchronously -> rejects and logs unexpe
cted error (6 ms)
    √ uploadBufferToCloudinary - unexpected error fallback message when thrown value lack
s message (4 ms)
    √ module import warns for any missing Cloudinary credential combinations (10 ms)

Test Suites: 1 passed, 1 total
Tests:       25 passed, 25 total
Snapshots:   0 total
Time:        2.848 s, estimated 3 s
```

- **Db:**

```
PS C:\Users\kakad\OneDrive\Desktop\Payment\BidSphere-Online-Auction-System\backend> npx jest .\test
 for_services\db.service.test.js --runInBand
 PASS  test_for_services/db.service.test.js
  connectDB (backend/services/db.js)
    √ exports a default async function (23 ms)
    √ calls mongoose.connect with MONGO_URI when set (8 ms)
    √ calls mongoose.connect with undefined when MONGO_URI not set (6 ms)
    √ invokes mongoose.connect only once per call (6 ms)
    √ handles synchronous throw from mongoose.connect and logs 'config error' (5 ms)
    √ resolves when mongoose.connect returns a resolved promise (5 ms)
    √ resolves when mongoose.connect returns non-promise value (3 ms)
    √ works with long MONGO_URI values (5 ms)
    √ multiple sequential calls call mongoose.connect repeatedly (4 ms)
    √ does not call console.log on successful connect (4 ms)
    √ uses current process.env value at call time (reflects changes) (5 ms)
    √ returns quickly even if mongoose.connect returns a delayed promise (3 ms)
    √ calling connectDB when mongoose.connect throws different errors still logs once (3 ms)
    √ does not produce unhandled rejection when mongoose.connect returns resolved promise (3 ms)
    √ gracefully handles being called with no side effects repeatedly (3 ms)

Test Suites: 1 passed, 1 total
Tests:       15 passed, 15 total
Snapshots:   0 total
Time:        2.222 s, estimated 3 s
Ran all test suites matching .\test_for_services\db.service.test.js.
```

- **Email_sender:**

```
PASS test_for_services/email.sender.test.js
  email.sender service
    √ SendVerificationCode - sends POST with verification code (105 ms)
    √ SendVerificationCode - fetch rejects -> function catches and logs (14 ms)
    √ WelcomeEmail - sends welcome email with name replaced (15 ms)
    √ WelcomeEmail - fetch error is swallowed and logged (13 ms)
    √ SendOutBidEmail - replaces placeholders and calls fetch with correct subject (16 ms
)
    √ SendResetPwdEmail - includes reset link in htmlContent (14 ms)
    √ SendAuctionWinnerEmail - includes winner name and auctionName (13 ms)
    √ SendCODSelectedEmail - includes name and auctionName (14 ms)
    √ SendUPISelectedEmail - generates QR, includes base64 attachment (16 ms)
    √ SendUPISelectedEmail - QRCode failure is caught and logged (13 ms)
    √ SendOutBidEmail - fetch rejection is caught and logged (9 ms)
    √ SendResetPwdEmail - fetch rejection is caught and logged (10 ms)
    √ SendAuctionWinnerEmail - fetch rejection is caught and logged (14 ms)
    √ SendCODSelectedEmail - fetch rejection is caught and logged (9 ms)
    √ SendPaymentVerifiedEmail - fetch rejection is caught and logged (12 ms)
    √ SendPaymentRejection - fetch rejection is caught and logged (9 ms)
    √ SendPaymentVerificationRequestSent - fetch rejection is caught and logged (10 ms)
    √ SendSellerWinnerNotification - success when template present (14 ms)
    √ SendSellerWinnerNotification - another success path (fresh import) (18 ms)
    √ SendPaymentVerifiedEmail - sends payment verified email (11 ms)
    √ SendPaymentRejection - sends rejection reason (11 ms)
    √ SendPaymentVerificationRequestSent - includes reqFor replacement (10 ms)
    √ SendSellerWinnerNotification - missing template triggers catch and logs error (9 ms
)
    √ multiple functions include BREVO api-key header in fetch (14 ms)

Test Suites: 1 passed, 1 total
Tests:       24 passed, 24 total
Snapshots:   0 total
Time:        2.855 s
Ran all test suites matching test_for_services/email.sender.test.js.
```

- **GetUser:**

```
PS C:\Users\kakad\OneDrive\Desktop\Payment\BidSphere-Online-Auction-System\backend> npx jest c:\Users\kakad\OneDrive\Desktop\Payment\B
idSphere-Online-Auction-System\backend\test_for_services\getUser.test.js
 PASS  test_for_services/getUser.test.js
  setUser
    √ signs a JWT with _id and email (10 ms)
    √ returns whatever jwt.sign returns (2 ms)
    √ uses secret '$@BidSphere' exactly (3 ms)
    √ ignores extra fields on user when building token payload (2 ms)
    √ works if email is null (still passes email: null) (3 ms)
    √ works if email is undefined (still passes email: undefined) (3 ms)
    √ throws if jwt.sign throws (65 ms)
    √ supports numeric-like string id (2 ms)
    √ supports non-string id (e.g., ObjectId-like) (2 ms)
    √ allows empty string email (2 ms)
    √ allows empty string _id (2 ms)
    √ still works even if user object has no _id and no email (1 ms)
  getUser
    √ returns null if token is null (5 ms)
    √ returns null if token is undefined (1 ms)
    √ returns null if token is empty string (1 ms)
    √ returns null if jwt.verify throws generic error (3 ms)
    √ returns null if jwt.verify throws TokenExpiredError-like (2 ms)
    √ returns null if payload is null (1 ms)
    √ returns null if payload is not an object (2 ms)
    √ returns null if payload._id is missing (1 ms)
    √ returns null if payload._id is null (3 ms)
    √ returns null if payload._id is empty string (1 ms)
    √ calls jwt.verify with secret '$@BidSphere' (2 ms)
    √ calls User.findById with payload._id (1 ms)
    √ uses correct projection when selecting user fields (1 ms)
    √ returns user object when found (1 ms)
    √ returns null if user not found (1 ms)
    √ propagates error if User.findById throws BEFORE select is accessed (4 ms)
    √ propagates error if .select() rejects (2 ms)
    √ handles case where jwt.verify returns payload with extra fields (2 ms)
    √ works when payload._id is numeric (2 ms)
    √ can be called multiple times and reuses mocks (1 ms)
    √ does not modify the token string passed in (1 ms)
    √ jwt.verify is only called once per getUser call (1 ms)
    √ does not call User.findById when token invalid (2 ms)
    √ handles token that yields payload with only _id (1 ms)
  generateHashPassword
    √ calls bcrypt.hash with the password and salt rounds 10 (5 ms)
```

```
    √ can be called multiple times and reuses mocks (1 ms)
    √ does not modify the token string passed in (1 ms)
    √ jwt.verify is only called once per getUser call (1 ms)
    √ does not call User.findById when token invalid (2 ms)
    √ handles token that yields payload with only _id (1 ms)
  generateHashPassword
    √ calls bcrypt.hash with the password and salt rounds 10 (5 ms)
    √ returns the hash value from bcrypt.hash (1 ms)
    √ can hash different passwords independently (1 ms)
    √ propagates error when bcrypt.hash rejects (2 ms)
    √ allows empty string password (1 ms)
    √ allows very long password (1 ms)

Test Suites: 1 passed, 1 total
Tests:       42 passed, 42 total
Snapshots:   0 total
Time:        2.579 s
Ran all test suites matching c:\Users\kakad\OneDrive\Desktop\Payment\BidSphere-Online-Auction-System\backend\test_for_services\getUser
.test.js.
PS C:\Users\kakad\OneDrive\Desktop\Payment\BidSphere-Online-Auction-System\backend>
```

- # Logger_service:

```
PS C:\Users\kakad\OneDrive\Desktop\Payment\BidSphere-Online-Auction-System\backend> npx jest .\test
_for_services\logger.service.test.js --runInBand
 PASS  test_for_services/logger.service.test.js
  logAuctionEvent
    √ calls AuctionLog.findOneAndUpdate with provided fields and upsert true (14 ms)
    √ uses empty object for default details when not provided (7 ms)
    √ accepts an explicit empty details object (2 ms)
    √ accepts null details and pushes null (2 ms)
    √ works without userName (undefined) (3 ms)
    √ works without userId (undefined) (3 ms)
    √ handles multiple calls and different auctionIds (3 ms)
    √ does not throw and logs error when AuctionLog.findOneAndUpdate throws (4 ms)
    √ always sets upsert true in options (2 ms)
    √ preserves the provided type string (3 ms)
    √ returns undefined on success (1 ms)
    √ accepts falsy auctionId and still calls findOneAndUpdate (2 ms)
    √ does not call console.error on successful write (3 ms)

Test Suites: 1 passed, 1 total
Tests:       13 passed, 13 total
Snapshots:   0 total
Time:        2.287 s
Ran all test suites matching .\test_for_services\logger.service.test.js.
PS C:\Users\kakad\OneDrive\Desktop\Payment\BidSphere-Online-Auction-System\backend>
```

- # Payment

```
 PASS  test_for_services/payment.service.test.js
  generateUpiLink
    √ throws when auction not found (36 ms)
    √ returns a valid upi link for integer registration fee (2 ms)
    √ returns link for decimal registration fee (3 ms)
    √ handles zero registration fee (3 ms)
    √ handles negative registration fee (still stringified) (2 ms)
    √ uses PAYMENT_UPI_ID env var in pa param (3 ms)
    √ calls Auction.findById with provided auctionId (4 ms)
    √ handles registrationFees passed as string (3 ms)
    √ encodes special characters in PAYMENT_UPI_ID (2 ms)
    √ works for very large registrationFees (2 ms)
    √ returns only expected query keys (2 ms)
    √ includes IN R currency code exactly 'INR' (2 ms)
    √ reflects updated env var between calls (2 ms)
    √ behaves when PAYMENT_UPI_ID is missing (pa becomes 'undefined') (1 ms)
    √ always returns a string starting with 'upi://pay?' (2 ms)

Test Suites: 1 passed, 1 total
Tests:       15 passed, 15 total
Snapshots:   0 total
Time:        3.294 s
Ran all test suites matching .\test_for_services\payment.service.test.js.
PS C:\Users\kakad\OneDrive\Desktop\Payment\BidSphere-Online-Auction-System\backend>
```

- # CoverageReport of Service:

All files services

100% Statements 194/194 **100%** Branches 61/61 **100%** Functions 26/26 **100%** Lines 188/188

Press *n* or *j* to go to the next uncovered block, *b*, *p* or *k* for the previous block.

Filter:

File	Statements		Branches		Functions		Lines	
auth.js	100%	12/12	100%	6/6	100%	3/3	100%	10/10
autoBid.service.js	100%	51/51	100%	24/24	100%	2/2	100%	48/48
cloudinary.service.js	100%	38/38	100%	28/28	100%	7/7	100%	37/37
db.js	100%	4/4	100%	0/0	100%	1/1	100%	4/4
email.sender.js	100%	77/77	100%	0/0	100%	11/11	100%	77/77
logger.service.js	100%	4/4	100%	1/1	100%	1/1	100%	4/4
payment.service.js	100%	8/8	100%	2/2	100%	1/1	100%	8/8

Code coverage generated by istanbul at 2025-11-22T17:43:27.161Z

```
PS C:\Users\kakad\OneDrive\Desktop\Payment\BidSphere-Online-Auction-System\backend> npx jest --coverage --coverageReporters=text --runInBand --silent
 PASS  test_for_services/cloudinary.service.test.js
 PASS  test_for_services/email.sender.test.js
 PASS  test_for_services/auth.test.js
 PASS  test_for_services/autoBid.test.js
 PASS  test_for_services/db.service.test.js
 PASS  test_for_services/payment.service.test.js
 PASS  test_for_services/logger.service.test.js
----------------------------|---------|----------|---------|---------|-------------------
File                        | % Stmts | % Branch | % Funcs | % Lines | Uncovered Line #s
----------------------------|---------|----------|---------|---------|-------------------
All files                   |     100 |      100 |     100 |     100 |
 email-templates            |     100 |      100 |     100 |     100 |
  ...tion_email.template.js |     100 |      100 |     100 |     100 |
 services                   |     100 |      100 |     100 |     100 |
  auth.js                   |     100 |      100 |     100 |     100 |
  autoBid.service.js        |     100 |      100 |     100 |     100 |
  cloudinary.service.js     |     100 |      100 |     100 |     100 |
  db.js                     |     100 |      100 |     100 |     100 |
  email.sender.js           |     100 |      100 |     100 |     100 |
  logger.service.js         |     100 |      100 |     100 |     100 |
  payment.service.js        |     100 |      100 |     100 |     100 |
----------------------------|---------|----------|---------|---------|-------------------

Test Suites: 7 passed, 7 total
Tests:       148 passed, 148 total
Snapshots:   0 total
Time:        4.374 s, estimated 7 s
```

Frontend Unit Testing

Pages:

- **About:**

```
 ✓ src/pages/__tests__/About.test.jsx (1 test) 75ms
   ✓ About page (1)
     ✓ renders the core sections 74ms

 Test Files  1 passed (1)
      Tests  1 passed (1)
   Start at  09:11:51
   Duration  1.38s (transform 59ms, setup 153ms, collect 328ms, tests 75ms, environment 6
17ms, prepare 17ms)
```

- **Admin_Dashboard:**

```
 ✓ src/pages/__tests__/AdminDashboard.test.jsx (2 tests) 201ms
   ✓ AdminDashboard page (2)
     ✓ loads auctions and opens detail modal on row click 126ms
     ✓ verifies auction after confirmation 73ms

 Test Files  1 passed (1)
      Tests  2 passed (2)
   Start at  09:12:38
   Duration  1.45s (transform 116ms, setup 138ms, collect 373ms, tests 201ms, environment
540ms, prepare 16ms)
```

- **Admin_login:**

```
 ✓ src/pages/__tests__/AdminLogin.test.jsx (5 tests) 158ms
   ✓ AdminLogin page (5)
     ✓ requires email and password before submitting 79ms
     ✓ validates email format 22ms
     ✓ enforces password length 22ms
     ✓ submits credentials and navigates on success 16ms
     ✓ shows error toast on failed login and resets form 16ms

 Test Files  1 passed (1)
      Tests  5 passed (5)
   Start at  09:12:57
   Duration  1.40s (transform 99ms, setup 135ms, collect 377ms, tests 158ms, environment
536ms, prepare 16ms)
```

- **Auction_page:**

```
 ✓ src/pages/__tests__/Auctions.test.jsx (2 tests) 157ms
   ✓ Auctions page (2)
     ✓ fetches auctions with search query and renders cards 68ms
     ✓ applies filters client-side 87ms

 Test Files  1 passed (1)
      Tests  2 passed (2)
   Start at  09:13:33
   Duration  1.39s (transform 98ms, setup 134ms, collect 372ms, tests 157ms, environment
540ms, prepare 16ms)
```

- **Auction**

```
 ✓ src/pages/__tests__/AuctionSettings.test.jsx (1 test) 113ms
   ✓ AuctionSettings page (1)
     ✓ loads user data and submits updates 112ms

 Test Files  1 passed (1)
      Tests  1 passed (1)
   Start at  09:14:05
   Duration  1.32s (transform 91ms, setup 134ms, collect 344ms, tests 113ms, environment
534ms, prepare 16ms)
```

- **Auth:**

```
 ✓ src/pages/__tests__/Auth.test.jsx (4 tests) 127ms
   ✓ OtpPage (Auth.jsx) (4)
     ✓ renders input and buttons 69ms
     ✓ shows success toast and navigates when OTP is correct 21ms
     ✓ shows invalid OTP message when OTP is incorrect 20ms
     ✓ resend button triggers info toast 16ms

 Test Files  1 passed (1)
      Tests  4 passed (4)
   Start at  09:14:34
   Duration  1.33s (transform 60ms, setup 133ms, collect 333ms, tests 127ms, environment
547ms, prepare 16ms)
```

- **AuthFlow:**

```
 ✓ src/pages/__tests__/AuthFlows.test.jsx (1 test) 2ms
   ✓ AuthFlows page tests (pending) (1)
     ✓ is marked for future implementation 1ms

 Test Files  1 passed (1)
      Tests  1 passed (1)
   Start at  09:15:01
   Duration  900ms (transform 31ms, setup 133ms, collect 26ms, tests 2ms, environment 544
ms, prepare 16ms)
```

- **BidHistory:**

```
 ✓ src/pages/__tests__/BidHistory.test.jsx (2 tests) 147ms
   ✓ BidHistory page (2)
     ✓ renders bid table and summary stats after loading 103ms
     ✓ navigates back to auction page when back button is clicked 42ms

 Test Files  1 passed (1)
      Tests  2 passed (2)
   Start at  09:15:44
   Duration  1.39s (transform 92ms, setup 136ms, collect 365ms, tests 147ms, environment
544ms, prepare 17ms)
```

- **Categories:**

```
 ✓ src/pages/__tests__/Categories.test.jsx (2 tests) 140ms
   ✓ Categories page (2)
     ✓ filters categories by search query 91ms
     ✓ loads auctions for selected category 46ms

 Test Files  1 passed (1)
      Tests  2 passed (2)
   Start at  09:16:00
   Duration  1.46s (transform 118ms, setup 137ms, collect 435ms, tests 140ms, environment
546ms, prepare 16ms)
```

- **Contact:**

```
 ✓ src/pages/__tests__/Contact.test.jsx (1 test) 67ms
   ✓ Contact page (1)
     ✓ shows success toast when sending message 65ms

 Test Files  1 passed (1)
      Tests  1 passed (1)
   Start at  09:16:15
   Duration  1.25s (transform 55ms, setup 135ms, collect 310ms, tests 67ms, environment 5
42ms, prepare 16ms)
```

- **CookiePolicy:**

```
 ✓ src/pages/__tests__/CookiePolicy.test.jsx (1 test) 134ms
   ✓ CookiePolicy page (1)
     ✓ renders hero content and support links with deterministic date 133ms

 Test Files  1 passed (1)
      Tests  1 passed (1)
   Start at  09:16:32
   Duration  1.45s (transform 85ms, setup 135ms, collect 437ms, tests 134ms, environment
548ms, prepare 17ms)
```

- **CreateAuction:**

```
Create Auction payload: {
  title: 'Rare Artifacts',
  name: 'Ancient Vase',
  description: 'Detailed description',
  images: [],
  category: 'art',
  condition: 'like-new',
  metadata: { conditionNotes: '' },
  startingPrice: 100,
  minIncrement: 10,
  startTime: '2025-01-01T03:30:00.000Z',
  endTime: '2025-01-05T06:30:00.000Z'
}

 ✓ src/pages/__tests__/CreateAuction.test.jsx (2 tests) 231ms
   ✓ CreateAuction page (2)
     ✓ enforces required field validation before submit 57ms
     ✓ submits form with valid payload and shows success modal 172ms

 Test Files  1 passed (1)
      Tests  2 passed (2)
   Start at  09:16:48
   Duration  1.47s (transform 113ms, setup 136ms, collect 365ms, tests 231ms, environment
 542ms, prepare 16ms)
```

- **Deliverycreate:**

```
 ✓ src/pages/__tests__/DeliveryCreate.test.jsx (2 tests) 115ms
   ✓ DeliveryCreate page (2)
     ✓ prefills address fields from current user and shows auction title 56ms
     ✓ submits delivery details and navigates to buyer dashboard 58ms

 Test Files  1 passed (1)
      Tests  2 passed (2)
   Start at  09:17:11
   Duration  1.35s (transform 87ms, setup 136ms, collect 371ms, tests 115ms, environment
535ms, prepare 16ms)
```

- **EditAuctionDraft:**

```
 ✓ src/pages/__tests__/EditAuctionDraft.test.jsx (1 test) 168ms
   ✓ EditAuctionDraft page (1)
     ✓ loads existing auction data and submits an update 166ms

 Test Files  1 passed (1)
      Tests  1 passed (1)
   Start at  09:17:31
   Duration  1.43s (transform 110ms, setup 136ms, collect 391ms, tests 168ms, environment
 539ms, prepare 16ms)
```

- **ExploreCategories:**

```
 ✓ src/pages/__tests__/ExploreCategories.test.jsx (2 tests) 106ms
   ✓ ExploreCategories section (2)
     ✓ renders default category links with navigation targets 94ms
     ✓ falls back to placeholder image when category preview fails to load 10ms

 Test Files  1 passed (1)
      Tests  2 passed (2)
   Start at  09:17:49
   Duration  1.35s (transform 81ms, setup 135ms, collect 361ms, tests 106ms, environment
548ms, prepare 16ms)
```

- **Feedback:**

```
 ✓ src/pages/__tests__/Feedback.test.jsx (2 tests) 144ms
   ✓ Feedback page (2)
     ✓ sends feedback and clears the form on success 110ms
     ✓ shows an error hint when the request fails 33ms

 Test Files  1 passed (1)
      Tests  2 passed (2)
   Start at  09:18:07
   Duration  1.32s (transform 64ms, setup 135ms, collect 306ms, tests 144ms, environment
534ms, prepare 16ms)
```

- **ForgotPassword:**

```
 ✓ src/pages/__tests__/ForgotPassword.test.jsx (3 tests) 109ms
   ✓ ForgotPassword page (3)
     ✓ requires an email before submitting 69ms
     ✓ sends reset request and navigates back to login on success 18ms
     ✓ shows an error toast when the API call fails 20ms

 Test Files  1 passed (1)
      Tests  3 passed (3)
   Start at  09:18:42
   Duration  1.36s (transform 96ms, setup 135ms, collect 374ms, tests 109ms, environment
545ms, prepare 16ms)
```

- **Home(page):**

```
✓ src/pages/__tests__/Home.test.jsx (12 tests) 1257ms
  ✓ Home page (12)
    ✓ renders hero section with all elements  334ms
    ✓ fetches featured auctions and renders them as cards 197ms
    ✓ shows loading state while fetching featured auctions 50ms
    ✓ shows empty state when no featured auctions 70ms
    ✓ renders AuctionCard with different statuses 102ms
    ✓ renders AuctionCard with missing data gracefully 73ms
    ✓ renders how it works section 96ms
    ✓ renders testimonials section 89ms
    ✓ handles API errors gracefully 81ms
    ✓ cleans up useEffect on unmount 30ms
    ✓ renders featured auctions section header 62ms
    ✓ handles auction images correctly 67ms

 Test Files  1 passed (1)
      Tests  12 passed (12)
   Start at  23:19:01
   Duration  5.26s (transform 388ms, setup 444ms, collect 1.25s, tests 1.26s, environment
1.68s, prepare 65ms)
```

- **Help:**

```
 ✓ src/pages/__tests__/Help.test.jsx (3 tests) 233ms
   ✓ Help centre page (3)
     ✓ exposes quick actions and support links 111ms
     ✓ expands and collapses help categories on click 80ms
     ✓ shows FAQ answer when a question is toggled 41ms

  Test Files  1 passed (1)
       Tests  3 passed (3)
    Start at  09:18:58
    Duration  1.55s (transform 86ms, setup 135ms, collect 429ms, tests 233ms, environment
551ms, prepare 17ms)
```

- **ListingPages:**

```
 ✓ src/pages/__tests__/ListingsPages.test.jsx (1 test) 2ms
   ✓ Listings pages tests (pending) (1)
     ✓ awaits implementation 1ms

 Test Files  1 passed (1)
      Tests  1 passed (1)
   Start at  09:19:27
   Duration  916ms (transform 32ms, setup 145ms, collect 27ms, tests 2ms, environment 545
ms, prepare 16ms)
```

- **Login(Page):**

```
 ✓ src/pages/__tests__/Login.test.jsx (11 tests) 333ms
   ✓ Login page (11)
     ✓ renders and performs successful login (res.user present, getCurrentUser overwrites
) 99ms
     ✓ successful login without res.user and getCurrentUser fails (no localStorage), rese
ts form 30ms
     ✓ validates required fields 15ms
     ✓ validates email format 15ms
     ✓ validates password length 12ms
     ✓ shows error message on login failure and resets form 11ms
     ✓ shows default failure message when rejection has no message 23ms
     ✓ toggles remember checkbox and resets it after submit 19ms
     ✓ shows default success message when no message returned 13ms
     ✓ successful login where getCurrentUser returns valid response but no user (branch c
overage) 32ms
     ✓ successful login where getCurrentUser fails (triggers catch block), falls back to
loginUser data 63ms

 Test Files  1 passed (1)
      Tests  11 passed (11)
   Start at  09:19:41
   Duration  1.63s (transform 115ms, setup 134ms, collect 406ms, tests 333ms, environment
 553ms, prepare 17ms)
```

- **MyBigs:**

```
 ✓ src/pages/__tests__/MyBids.test.jsx (2 tests) 115ms
   ✓ MyBids page (2)
     ✓ loads bidding history and shows summary cards 63ms
     ✓ filters bids when switching tabs and using search 51ms

 Test Files  1 passed (1)
      Tests  2 passed (2)
   Start at  09:19:59
   Duration  1.36s (transform 104ms, setup 135ms, collect 376ms, tests 115ms, environment
 541ms, prepare 16ms)
```

- **Mylistings:**

```
 ✓ src/pages/__tests__/MyListings.test.jsx (2 tests) 121ms
   ✓ MyListings page (2)
     ✓ loads the seller listings and shows listing rows 92ms
     ✓ removes a listing when the seller confirms deletion 28ms

 Test Files  1 passed (1)
      Tests  2 passed (2)
   Start at  09:20:14
   Duration  1.36s (transform 91ms, setup 133ms, collect 363ms, tests 121ms, environment
548ms, prepare 16ms)
```

- **Payfees:**

```
 ✓ src/pages/__tests__/PayFees.test.jsx (2 tests) 150ms
   ✓ PayFees page (2)
     ✓ creates a UPI payment and submits verification details 132ms
     ✓ records a COD payment path when selected 17ms

 Test Files  1 passed (1)
      Tests  2 passed (2)
   Start at  09:20:30
   Duration  1.39s (transform 100ms, setup 133ms, collect 377ms, tests 150ms, environmen
534ms, prepare 16ms)
```

- **PrivacyPolicy:**

```
 ✓ src/pages/__tests__/PrivacyPolicy.test.jsx (1 test) 113ms
   ✓ PrivacyPolicy page (1)
     ✓ renders hero copy and contact links with deterministic date 112ms

 Test Files  1 passed (1)
      Tests  1 passed (1)
   Start at  09:20:48
   Duration  1.41s (transform 72ms, setup 134ms, collect 427ms, tests 113ms, environment
539ms, prepare 16ms)
```

- **RatingForm:**

```
 ✓ src/pages/__tests__/RatingForm.test.jsx (4 tests) 160ms
   ✓ RatingForm component (4)
     ✓ returns null if required ids missing 14ms
     ✓ submits a brand new rating 81ms
     ✓ updates an existing rating when already rated 26ms
     ✓ allows deleting an existing rating via summary card 39ms

 Test Files  1 passed (1)
      Tests  4 passed (4)
   Start at  09:21:07
   Duration  1.36s (transform 87ms, setup 132ms, collect 340ms, tests 160ms, environmen
538ms, prepare 16ms)
```

- **Register:**

```
 ✓ src/pages/__tests__/Register.test.jsx (8 tests) 208ms
   ✓ Register page (8)
     ✓ renders form fields and submits data to registerUser 97ms
     ✓ renders the register banner image 5ms
     ✓ validates required fields 25ms
     ✓ validates email format 16ms
     ✓ validates password length 15ms
     ✓ shows alert on registerUser failure and resets form 16ms
     ✓ shows default alert message on success when no message is returned 15ms
     ✓ shows default alert message on failure when no message is returned 16ms

 Test Files  1 passed (1)
      Tests  8 passed (8)
   Start at  09:21:19
   Duration  1.46s (transform 96ms, setup 134ms, collect 373ms, tests 208ms, environment
542ms, prepare 16ms)
```

- **ResetPassword:**

```
 ✓ src/pages/__tests__/ResetPassword.test.jsx (4 tests) 128ms
   ✓ ResetPassword page (4)
     ✓ prefills email from URL param and requires all fields 75ms
     ✓ validates matching passwords and length 28ms
     ✓ calls resetPassword and navigates on success 11ms
     ✓ shows default error when API rejects without message 12ms

 Test Files  1 passed (1)
      Tests  4 passed (4)
   Start at  09:21:32
   Duration  1.36s (transform 89ms, setup 132ms, collect 367ms, tests 128ms, environment
538ms, prepare 16ms)
```

- **SellerInfo:**

```
 ✓ src/pages/__tests__/SellerInfo.test.jsx (3 tests) 84ms
   ✓ SellerInfo page (3)
     ✓ shows loading state then seller header and empty reviews 44ms
     ✓ displays error when seller id invalid 7ms
     ✓ renders reviews list when data available 32ms

 Test Files  1 passed (1)
      Tests  3 passed (3)
   Start at  09:21:46
   Duration  1.33s (transform 94ms, setup 134ms, collect 378ms, tests 84ms, environment 5
44ms, prepare 16ms)
```

- **SellerPages:**

```
 ✓ src/pages/__tests__/SellerPages.test.jsx (1 test) 2ms
   ✓ Seller pages tests (pending) (1)
     ✓ awaits real tests 1ms

 Test Files  1 passed (1)
      Tests  1 passed (1)
   Start at  09:21:58
   Duration  914ms (transform 32ms, setup 135ms, collect 27ms, tests 2ms, environment 55
ms, prepare 17ms)

 PASS  Waiting for file changes...
```

- **SellerRating:**

```
 ✓ src/pages/__tests__/SellerRating.test.jsx (4 tests) 102ms
   ✓ SellerRating component (4)
     ✓ returns null when no sellerId provided 14ms
     ✓ loads ratings and shows averages 49ms
     ✓ shows no ratings when API returns empty 21ms
     ✓ shows error message if API fails 15ms

 Test Files  1 passed (1)
      Tests  4 passed (4)
   Start at  09:22:11
   Duration  1.31s (transform 86ms, setup 134ms, collect 338ms, tests 102ms, environment
540ms, prepare 16ms)
```

- **StaticPages:**

```
 ✓ src/pages/__tests__/StaticPages.test.jsx (1 test) 2ms
   ✓ Static pages tests (pending) (1)
     ✓ awaits coverage 1ms

 Test Files  1 passed (1)
      Tests  1 passed (1)
   Start at  09:22:34
   Duration  898ms (transform 32ms, setup 134ms, collect 26ms, tests 2ms, environment 541
ms, prepare 16ms)
```

- **TermsOfService:**

```
 ✓ src/pages/__tests__/TermsOfService.test.jsx (2 tests) 138ms
   ✓ TermsOfService page (2)
     ✓ renders hero section with last updated date and CTA links 114ms
     ✓ lists key policy sections and bullet items 22ms

 Test Files  1 passed (1)
      Tests  2 passed (2)
   Start at  09:22:46
   Duration  1.39s (transform 82ms, setup 136ms, collect 357ms, tests 138ms, environment
553ms, prepare 20ms)
```

- **UserDashboardBuyer:**

```
 ✓ src/pages/__tests__/UserDashboardBuyer.test.jsx (3 tests) 240ms
   ✓ UserDashboardBuyer page (3)
     ✓ loads dashboard data and displays watchlist, unpaid wins, and deliveries 125ms
     ✓ submits delivery address for a paid auction without saved delivery 88ms
     ✓ downloads invoice when delivery exists 26ms

 Test Files  1 passed (1)
      Tests  3 passed (3)
   Start at  09:23:06
   Duration  1.53s (transform 127ms, setup 137ms, collect 410ms, tests 240ms, environment
 541ms, prepare 16ms)
```

- **UserDashboards:**

```
 ✓ src/pages/__tests__/UserDashboards.test.jsx (1 test) 2ms
   ✓ User dashboard tests (pending) (1)
     ✓ awaits scenarios 1ms

 Test Files  1 passed (1)
      Tests  1 passed (1)
   Start at  09:23:22
   Duration  897ms (transform 31ms, setup 134ms, collect 25ms, tests 2ms, environment 542
ms, prepare 16ms)
```

- **UserDashboardSeller:**

```
 ✓ src/pages/__tests__/UserDashboardSeller.test.jsx (3 tests) 200ms
   ✓ UserDashboardSeller page (3)
     ✓ loads seller profile, stats, and listings 70ms
     ✓ filters to ended listings when Earnings shortcut clicked 92ms
     ✓ deletes an auction after confirmation 36ms

 Test Files  1 passed (1)
      Tests  3 passed (3)
   Start at  09:23:36
   Duration  1.46s (transform 107ms, setup 135ms, collect 392ms, tests 200ms, environment
 538ms, prepare 16ms)
```

- **VerifyMail:**

```
 ✓ src/pages/__tests__/VerifyMail.test.jsx (8 tests) 196ms
   ✓ VerifyEmail page (8)
     ✓ prefills email from location.state 32ms
     ✓ validates missing email/code and shows toast error 51ms
     ✓ calls verifyEmail and navigates on success 21ms
     ✓ shows error toast on verifyEmail rejection 26ms
     ✓ shows loading state while verifyEmail is pending 16ms
     ✓ resend shows instructional toast info 6ms
     ✓ shows default success message when verifyEmail returns no message 26ms
     ✓ shows default failure message when reject has no message 16ms

 Test Files  1 passed (1)
      Tests  8 passed (8)
   Start at  09:23:48
   Duration  1.46s (transform 86ms, setup 136ms, collect 371ms, tests 196ms, environment
557ms, prepare 16ms)
```

- **Watchlist:**

```
Auction ID: a1
Auction object: {
  _id: 'a1',
  title: 'Vintage Guitar',
  currentBid: 5000,
  startingPrice: 3000,
  endTime: '2025-01-01T00:00:00Z',
  totalBids: 7,
  item: { category: 'Music', condition: 'Used', images: [] }
}

 ✓ src/pages/__tests__/Watchlist.test.jsx (4 tests) 173ms
   ✓ Watchlist page (4)
     ✓ loads watchlist entries and shows summary counts 102ms
     ✓ filters when search query entered 28ms
     ✓ removes item after confirmation 20ms
     ✓ shows empty state when API returns empty list 21ms

 Test Files  1 passed (1)
      Tests  4 passed (4)
   Start at  09:24:02
   Duration  1.42s (transform 94ms, setup 137ms, collect 370ms, tests 173ms, environmen
545ms, prepare 16ms)
```

- **AuctionDetails:**

```
✓ src/pages/__tests__/AuctionDetails.test.jsx (26 tests) 1678ms
  ✓ AuctionDetails page (26)
    ✓ renders basic auction details for unpaid users 267ms
    ✓ allows adding to watchlist from unpaid view 66ms
    ✓ allows removing from watchlist 83ms
    ✓ shows loading state while fetching auction 7ms
    ✓ shows error message when auction fetch fails 19ms
    ✓ shows auction not found when no auction data 23ms
    ✓ allows placing a bid when user is logged in and has paid 114ms
    ✓ shows error when bid is too low 88ms
    ✓ validates bid amount before submission 102ms
    ✓ toggles auto-bid functionality 146ms
    ✓ shows payment status for paid users 67ms
    ✓ displays seller rating component 20ms
    ✓ displays countdown timer for live auction 21ms
    ✓ shows ended message when auction has ended 57ms
    ✓ shows bid history when available 65ms
    ✓ handles share functionality 44ms
    ✓ handles report functionality 43ms
    ✓ shows seller view for seller users 20ms
    ✓ handles different auction statuses 77ms
    ✓ covers edge cases for image handling 49ms
    ✓ handles auto-bid edit scenario 59ms
    ✓ covers timer edge cases 33ms
    ✓ handles auction ended with winner info 36ms
    ✓ handles registered user payment status 31ms
    ✓ handles clipboard fallback for share 103ms
    ✓ handles mailto fallback for share 32ms

Test Files  1 passed (1)
     Tests  26 passed (26)
  Start at  22:47:19
  Duration  5.32s (transform 447ms, setup 366ms, collect 1.14s, tests 1.68s, environment
1.58s, prepare 46ms)
```

- **Covarge_report:**

% Coverage report from v8

File	% Stmts	% Branch	% Funcs	% Lines	Uncovered Line #s
All files	61.58	51.22	58.66	66.47	
src	0	100	0	0	
App.jsx	0	100	0	0	39-46
main.jsx	0	100	100	0	9-10
src/api	0	0	0	0	
index.js	0	0	0	0	2-225
src/components	0	0	0	0	
Footer.jsx	0	100	0	0	5
Navbar.jsx	0	0	0	0	8-165
SellerRatingSummary.jsx	0	0	0	0	6-60
src/constants	0	100	100	0	
categories.js	0	100	100	0	2
src/pages	70.1	55.3	68.85	75.05	
About.jsx	100	100	100	100	
AdminDashboard.jsx	67.36	46.96	66	72.12	...,458-493,570,593
AdminLogin.jsx	94.28	77.27	100	100	27,34,46-53
AuctionDetails.jsx	71.27	66.22	69.09	73.88	...4-1029,1061,1106
AuctionSettings.jsx	71.42	47.82	81.81	73.46	...2-85,105-106,148
Auctions.jsx	85	78.02	84.21	91.13	...,131,179-180,187
Auth.jsx	100	100	100	100	
BidHistory.jsx	88.09	71.79	70	89.74	22,65-82
Categories.jsx	81.33	60.91	88.88	82.35	...1-93,116,120-127
CookiePolicy.jsx	100	100	100	100	
CreateAuction.jsx	41.17	42.75	34	48.73	...,399-478,575-616
DeliveryCreate.jsx	84.61	58.73	80	90	47,58,74-78,99
EditAuctionDraft.jsx	36.48	38.46	25.45	47.02	...,384-396,426-540
ExploreCategories.jsx	100	100	100	100	
Feedback.jsx	100	100	100	100	
ForgotPassword.jsx	100	75	100	100	17-20
Help.jsx	100	85.71	100	100	14-121
Home.jsx	95.23	87.23	100	94.73	31,42
Login.jsx	100	100	100	100	
MyBids.jsx	80	55.55	65.38	85.54	...,278-279,338-450
MyListings.jsx	86.11	55.55	100	90	53,69-70
PayFees.jsx	71.96	57.25	87.5	78.26	...-130,134-141,201
PrivacyPolicy.jsx	100	100	100	100	
RatingForm.jsx	82.95	72.09	84.61	83.95	...-123,139,168-169
Register.jsx	100	100	100	100	
RegistrationFee.jsx	0	0	0	0	13-170
ResetPassword.jsx	100	80	100	100	17-18,41-45
SellerInfo.jsx	84	57.4	81.81	88.88	44,57-59,95
SellerRating.jsx	88.09	61.81	100	91.66	31-33
TermsOfService.jsx	100	100	100	100	
UserDashboardBuyer.jsx	78.16	42.95	72.91	79.37	...,599,615,666-667
UserDashboardSeller.jsx	84.09	64.13	85.71	86.11	...,208-223,250-251
VerifyMail.jsx	100	100	100	100	
Watchlist.jsx	75.67	43.67	92.85	82.53	...-138,151,171-172
contact.jsx	100	100	100	100	
src/utils	12.72	8.33	40	13.2	
invoicePDF.js	0	0	0	0	5-184
statusHelpers.js	46.66	37.5	100	53.84	5,9,14-16,24-26

Api:

All files src/api

100% Statements 164/164 **100%** Branches 64/64 **100%** Functions 62/62 **100%** Lines 118/118

Press *n* or *j* to go to the next uncovered block, *b*, *p* or *k* for the previous block.

Filter:

File	Statements		Branches		Functions		Lines	
index.js	100%	164/164	100%	64/64	100%	62/62	100%	118/118

```
 ✓ src/api/__tests__/index.test.js (79 tests) 129ms
   ✓ POST helpers route to correct endpoints (27)
     ✓ { label: 'registerUser', invoke: [Function invoke], expectedPath: 'http://localhost:5000/b
idsphere/user/register', expectedBody: { username: 'demo', email: 'demo@test.com' } } uses postJS
ON helper 26ms
     ✓ { label: 'loginUser', invoke: [Function invoke], expectedPath: 'http://localhost:5000/bids
phere/user/login', expectedBody: { email: 'me@test.com', password: 'secret' } } uses postJSON hel
per 3ms
     ✓ { label: 'logoutUser', invoke: [Function invoke], expectedPath: 'http://localhost:5000/bid
sphere/user/logout', expectedBody: {} } uses postJSON helper 2ms
     ✓ { label: 'verifyEmail', invoke: [Function invoke], expectedPath: 'http://localhost:5000/bi
dsphere/user/verifyemail', expectedBody: { email: 'a@b.com', code: '123456' } } uses postJSON hel
per 2ms
     ✓ { label: 'loginAdmin', invoke: [Function invoke], expectedPath: 'http://localhost:5000/bid
sphere/admin/login', expectedBody: { email: 'admin@test.com', password: 'pass' } } uses postJSON
helper 2ms
     ✓ { label: 'logoutAdmin', invoke: [Function invoke], expectedPath: 'http://localhost:5000/bi
dsphere/admin/logout', expectedBody: {} } uses postJSON helper 2ms
     ✓ { label: 'verifyAuction', invoke: [Function invoke], expectedPath: 'http://localhost:5000/
bidsphere/admin/auctions/auction-123/verify', expectedBody: {} } uses postJSON helper 1ms
     ✓ { label: 'removeAuctionAdmin', invoke: [Function invoke], expectedPath: 'http://localhost:
5000/bidsphere/admin/auctions/auction-123/remove', expectedBody: {} } uses postJSON helper 2ms
     ✓ { label: 'createAuction', invoke: [Function invoke], expectedPath: 'http://localhost:5000/
bidsphere/auctions/create', expectedBody: { title: 'Item' } } uses postJSON helper 4ms
     ✓ { label: 'addToWatchlist', invoke: [Function invoke], expectedPath: 'http://localhost:5000
/bidsphere/user/watchlist', expectedBody: { auctionId: 'auction-123' } } uses postJSON helper 2ms
     ✓ { label: 'uploadImagesBase64', invoke: [Function invoke], expectedPath: 'http://localhost:
5000/bidsphere/auctions/upload-base64', expectedBody: { images: [ 'base64' ] } } uses postJSON he
lper 2ms
     ✓ { label: 'placeBid', invoke: [Function invoke], expectedPath: 'http://localhost:5000/bidsp
here/auctions/auction-123/bid/place', expectedBody: { amount: 400 } } uses postJSON helper 1ms
     ✓ { label: 'setAutoBid', invoke: [Function invoke], expectedPath: 'http://localhost:5000/bid
sphere/auctions/auction-123/bid/setauto', expectedBody: { maxLimit: 1000 } } uses postJSON helper
 4ms
```

```
    ✓ { label: 'editAutoBid', invoke: [Function invoke], expectedPath: 'http://localhost:5000/bi
dsphere/auctions/auction-123/bid/editauto/auto-999', expectedBody: { maxLimit: 900 } } uses postJ
SON helper 3ms
    ✓ { label: 'activateAutoBid', invoke: [Function invoke], expectedPath: 'http://localhost:500
0/bidsphere/auctions/auction-123/bid/activateauto/auto-999', expectedBody: {} } uses postJSON hel
per 2ms
    ✓ { label: 'deactivateAutoBid', invoke: [Function invoke], expectedPath: 'http://localhost:5
000/bidsphere/auctions/auction-123/bid/deactivateauto/auto-999', expectedBody: {} } uses postJSON
 helper 2ms
    ✓ { label: 'confirmAdminNotification', invoke: [Function invoke], expectedPath: 'http://loca
lhost:5000/bidsphere/admin/notifications/note-1/confirm', expectedBody: {} } uses postJSON helper
 1ms
    ✓ { label: 'rejectAdminNotification', invoke: [Function invoke], expectedPath: 'http://local
host:5000/bidsphere/admin/notifications/note-2/reject', expectedBody: {} } uses postJSON helper 1
ms
    ✓ { label: 'createRegistrationPayment', invoke: [Function invoke], expectedPath: 'http://loc
alhost:5000/bidsphere/auctions/auction-123/au-registration/pay', expectedBody: {} } uses postJSON
 helper 1ms
    ✓ { label: 'verifyAuctionPayment', invoke: [Function invoke], expectedPath: 'http://localhos
t:5000/bidsphere/auctions/auction-123/pay-777/verify', expectedBody: { approved: true } } uses po
stJSON helper 1ms
    ✓ { label: 'createWinningCodPayment', invoke: [Function invoke], expectedPath: 'http://local
host:5000/bidsphere/auctions/auction-123/finalpay/cod', expectedBody: {} } uses postJSON helper 1
ms
    ✓ { label: 'createWinningUpiPayment', invoke: [Function invoke], expectedPath: 'http://local
host:5000/bidsphere/auctions/auction-123/finalpay/upi', expectedBody: {} } uses postJSON helper 1
ms
    ✓ { label: 'createDelivery', invoke: [Function invoke], expectedPath: 'http://localhost:5000
/bidsphere/delivery/create', expectedBody: { address: '123 Main' } } uses postJSON helper 1ms
    ✓ { label: 'rateSeller', invoke: [Function invoke], expectedPath: 'http://localhost:5000/bid
sphere/ratings', expectedBody: { sellerId: 'seller-1', rating: 5 } } uses postJSON helper 1ms
```

```
    ✓ { label: 'requestPasswordReset', invoke: [Function invoke], expectedPath: 'http://localhos
t:5000/bidsphere/user/forgetpwd', expectedBody: { email: 'reset@test.com' } } uses postJSON helpe
r 1ms
    ✓ { label: 'resetPassword', invoke: [Function invoke], expectedPath: 'http://localhost:5000/
bidsphere/user/resetpwd', expectedBody: { email: 'reset@test.com', token: 'OTP', password: 'newpa
ss' } } uses postJSON helper 1ms
    ✓ falls back to an empty object payload when body missing 1ms
  ✓ GET helpers route to correct endpoints (17)
    ✓ { label: 'getAllAuctionsAdmin with params', invoke: [Function invoke], expectedPath: 'http
://localhost:5000/bidsphere/admin/auctions?status=pending' } uses getJSON helper 2ms
    ✓ { label: 'getAuctionDetailsAdmin', invoke: [Function invoke], expectedPath: 'http://localh
ost:5000/bidsphere/admin/auctions/auction-123' } uses getJSON helper 1ms
    ✓ { label: 'getAuction', invoke: [Function invoke], expectedPath: 'http://localhost:5000/bid
sphere/auctions/auction-123' } uses getJSON helper 1ms
    ✓ { label: 'getMyAuctions', invoke: [Function invoke], expectedPath: 'http://localhost:5000/
bidsphere/auctions/mine?page=2' } uses getJSON helper 1ms
    ✓ { label: 'listAuctions', invoke: [Function invoke], expectedPath: 'http://localhost:5000/b
idsphere/auctions?limit=5' } uses getJSON helper 1ms
    ✓ { label: 'getCurrentUser', invoke: [Function invoke], expectedPath: 'http://localhost:5000
/bidsphere/user/me' } uses getJSON helper 1ms
    ✓ { label: 'getUserById', invoke: [Function invoke], expectedPath: 'http://localhost:5000/bi
dsphere/user/user-88' } uses getJSON helper 1ms
    ✓ { label: 'getWatchlist', invoke: [Function invoke], expectedPath: 'http://localhost:5000/b
idsphere/user/watchlist?limit=10' } uses getJSON helper 1ms
    ✓ { label: 'getBiddingHistory', invoke: [Function invoke], expectedPath: 'http://localhost:5
000/bidsphere/user/bidding-history?page=1' } uses getJSON helper 1ms
    ✓ { label: 'getUserAutoBid', invoke: [Function invoke], expectedPath: 'http://localhost:5000
/bidsphere/auctions/auction-123/bid/myautobid' } uses getJSON helper 1ms
    ✓ { label: 'getAdminNotifications', invoke: [Function invoke], expectedPath: 'http://localho
st:5000/bidsphere/admin/notifications' } uses getJSON helper 1ms
    ✓ { label: 'getMyDeliveries', invoke: [Function invoke], expectedPath: 'http://localhost:500
0/bidsphere/delivery/my-deliveries' } uses getJSON helper 1ms
```

```
      ✓ { label: 'getAllDeliveries', invoke: [Function invoke], expectedPath: 'http://localhost:50
00/bidsphere/delivery/all?status=new' } uses getJSON helper 1ms
      ✓ { label: 'getMyPayments', invoke: [Function invoke], expectedPath: 'http://localhost:5000/
bidsphere/user/payments' } uses getJSON helper 1ms
      ✓ { label: 'getPayment', invoke: [Function invoke], expectedPath: 'http://localhost:5000/bid
sphere/auctions/auction-123/payment/pay-777' } uses getJSON helper 1ms
      ✓ { label: 'listPayments', invoke: [Function invoke], expectedPath: 'http://localhost:5000/b
idsphere/admin/payments/payments?status=pending' } uses getJSON helper 1ms
      ✓ { label: 'getSellerRatings', invoke: [Function invoke], expectedPath: 'http://localhost:50
00/bidsphere/ratings/seller/seller-1' } uses getJSON helper 1ms
    ✓ DELETE helpers route to correct endpoints (3)
      ✓ { label: 'deleteAuction', invoke: [Function invoke], expectedPath: 'http://localhost:5000/
bidsphere/auctions/auction-123' } uses DELETE helper 1ms
      ✓ { label: 'removeFromWatchlist', invoke: [Function invoke], expectedPath: 'http://localhost
:5000/bidsphere/user/watchlist/auction-123' } uses DELETE helper 1ms
      ✓ { label: 'deleteRating', invoke: [Function invoke], expectedPath: 'http://localhost:5000/b
idsphere/ratings/rating-55' } uses DELETE helper 1ms
    ✓ PATCH and PUT helpers (5)
      ✓ saves auction draft via PATCH 2ms
      ✓ updateAuction uses JSON branch when body is plain object 1ms
      ✓ updateAuction routes to putFormData when FormData payload provided 1ms
      ✓ updateRating uses putJSON helper 1ms
      ✓ updateAuction propagates putFormData failures 3ms
    ✓ Specialized helpers behavior (11)
      ✓ getCategories trims names and deduplicates categories 1ms
      ✓ getCategories defaults to limit=200 when opts omitted 1ms
      ✓ getCategories returns empty array when backend omits auctions 1ms
      ✓ uploadImagesFormData resolves data on success 2ms
      ✓ uploadImagesFormData throws informative error on failure 3ms
      ✓ uploadImagesFormData throws default message when backend omits error text 1ms
      ✓ uploadImagesFormData tolerates json parsing errors when response is ok 2ms
      ✓ updateUserProfile succeeds when backend returns ok 1ms
      ✓ updateUserProfile throws descriptive error when backend rejects 1ms
      ✓ updateUserProfile falls back to generic message when response lacks detail 1ms
```

```
      ✓ updateUserProfile uses default error when json parsing fails 1ms
    ✓ Helper error handling (7)
      ✓ postJSON propagates server payloads on failure 1ms
      ✓ patchJSON throws Error with message when response not ok 1ms
      ✓ getJSON throws server payload when response not ok 1ms
      ✓ putJSON throws when response not ok 1ms
      ✓ delete helper propagates server error payload 1ms
      ✓ patchJSON falls back to generic message when server omits message 1ms
      ✓ putJSON falls back to generic message when server omits message 1ms
    ✓ Helper success fallbacks (2)
      ✓ patchJSON resolves null when JSON parsing fails 1ms
      ✓ putJSON resolves null when JSON parsing fails 1ms
    ✓ Query string fallbacks (7)
      ✓ getAllAuctionsAdmin omits query string when params are empty 1ms
      ✓ getAllAuctionsAdmin omits query string when params are empty 1ms
      ✓ getMyAuctions omits query string when params are empty 1ms
      ✓ listAuctions omits query string when params are empty 1ms
      ✓ getWatchlist omits query string when params are empty 1ms
      ✓ getBiddingHistory omits query string when params are empty 1ms
      ✓ getAllDeliveries omits query string when params are empty 1ms
      ✓ listPayments omits query string when params are empty 1ms

  Test Files  1 passed (1)
       Tests  79 passed (79)
    Start at  23:36:11
    Duration  6.05s (transform 423ms, setup 631ms, collect 408ms, tests 129ms, environment 3.61s, prepare 92ms)

 PASS  Waiting for file changes...
       press h to show help, press q to quit
```

Components:

All files **src/components**

100% Statements 123/123 **100%** Branches 79/79 **100%** Functions 28/28 **100%** Lines 106/106

Press *n* or *j* to go to the next uncovered block, *b*, *p* or *k* for the previous block.

Filter:

File		Statements		Branches		Functions		Lines	
Footer.jsx		100%	1/1	100%	0/0	100%	1/1	100%	1/1
Navbar.jsx		100%	83/83	100%	46/46	100%	20/20	100%	72/72
SellerRatingSummary.jsx		100%	39/39	100%	33/33	100%	7/7	100%	33/33

```
✓ src/components/__tests__/Footer.test.jsx (2 tests) 407ms
  ✓ Footer component (2)
    ✓ shows bidder and seller sections 279ms
    ✓ renders policy links with correct targets 121ms

Test Files  1 passed (1)
     Tests  2 passed (2)
  Start at  23:41:06
  Duration  5.49s (transform 220ms, setup 601ms, collect 1.23s, tests 407ms, environment 2.40s, prepare 67ms)

PASS  Waiting for file changes...
      press h to show help, press q to quit
```

```
✓ src/components/__tests__/Navbar.test.jsx (20 tests) 2760ms
  ✓ Navbar (18)
    ✓ shows login/register links when no user/admin present  327ms
    ✓ navigates to search results when query submitted 299ms
    ✓ navigates to search results when search button clicked with text 273ms
    ✓ toggles user menu and logs out via logoutUser 245ms
    ✓ shows admin controls and logs out admins 291ms
    ✓ falls back to sessionStorage when localStorage access fails 30ms
    ✓ falls back to null when sessionStorage has no cached user 31ms
    ✓ falls back to null when both storages are unavailable 113ms
    ✓ gracefully handles malformed stored user JSON and storage events 51ms
    ✓ ignores storage events that target other keys 43ms
    ✓ closes the account menu when clicking outside 218ms
    ✓ navigates to auctions when search is blank and button pressed 97ms
    ✓ navigates to auctions when Enter is pressed on an empty search 106ms
    ✓ logs a warning when logoutUser rejects 214ms
    ✓ renders email initials when username missing and shows fallback name in the menu header 108ms
    ✓ falls back to an empty account label when both username and email are missing 110ms
    ✓ logs a warning when logoutAdmin rejects 180ms
    ✓ ignores late getCurrentUser responses after unmount 15ms
  ✓ runAdminLogoutSequence helper (2)
    ✓ skips API call when admin absent but still clears state 1ms
    ✓ logs a warning when logoutAdminFn rejects 1ms

 Test Files  1 passed (1)
      Tests  20 passed (20)
   Start at  23:39:10
   Duration  10.80s (transform 673ms, setup 897ms, collect 4.42s, tests 2.76s, environment 1.91s, prepare 66ms)

 PASS  Waiting for file changes...
       press h to show help, press q to quit
```

```
✓ src/components/__tests__/SellerRatingSummary.test.jsx (10 tests) 434ms
  ✓ SellerRatingSummary (10)
    ✓ returns null when sellerId is missing 67ms
    ✓ skips fetching when sellerId is not a valid ObjectId 32ms
    ✓ renders average rating and review count on success 116ms
    ✓ uses the ratings array fallback when data is missing 34ms
    ✓ falls back to an empty array when both data sources are missing 20ms
    ✓ shows an em dash and singular label when ratings average to zero 26ms
    ✓ shows an error message when the API fails 30ms
    ✓ falls back to a default error message when rejection lacks details 15ms
    ✓ prevents state updates after unmount 66ms
    ✓ ignores rejected promises after unmount 22ms

 Test Files  1 passed (1)
      Tests  10 passed (10)
   Start at  23:40:06
   Duration  6.61s (transform 542ms, setup 655ms, collect 2.09s, tests 434ms, environment 2.51s, prepare 73ms)

 PASS  Waiting for file changes...
       press h to show help, press q to quit
```

Utils:

```
 ✓ src/utils/__tests__/invoicePDF.test.js (4 tests) 703ms
   ✓ generateInvoicePDF (4)
     ✓ renders HTML to canvas and saves the PDF when successful  354ms
     ✓ cleans up the temporary node when canvas rendering fails 112ms
     ✓ fills in fallback placeholders when optional data is missing 126ms
     ✓ uses alternate data fields when available and renders item images 105ms

 Test Files  1 passed (1)
      Tests  4 passed (4)
   Start at  23:41:45
   Duration  5.47s (transform 346ms, setup 896ms, collect 1.14s, tests 703ms, environment 1.96s, prepare 57ms)

 PASS  Waiting for file changes...
       press h to show help, press q to quit
```

```
 ✓ src/utils/__tests__/statusHelpers.test.js (5 tests) 21ms
   ✓ statusHelpers (5)
     ✓ maps known statuses to color tokens 7ms
     ✓ falls back to neutral styles for unknown or missing statuses 1ms
     ✓ returns friendly labels for specific statuses 3ms
     ✓ returns raw status text for unmapped values 2ms
     ✓ returns N/A when status missing 2ms

 Test Files  1 passed (1)
      Tests  5 passed (5)
   Start at  23:42:28
   Duration  3.02s (transform 116ms, setup 423ms, collect 54ms, tests 21ms, environment 1.82s, prepare 66ms)

 PASS  Waiting for file changes...
       press h to show help, press q to quit
```

All files **src/utils**

100% Statements 55/55 **100%** Branches 70/70 **100%** Functions 5/5 **100%** Lines 53/53

Press *n* or *j* to go to the next uncovered block, *b*, *p* or *k* for the previous block.

Filter:

File	Statements		Branches		Functions		Lines	
invoicePDF.js	100%	40/40	100%	56/56	100%	3/3	100%	40/40
statusHelpers.js	100%	15/15	100%	14/14	100%	2/2	100%	13/13

App:

All files src

100% Statements 7/7 **100%** Branches 0/0 **100%** Functions 3/3 **100%** Lines 7/7

Press *n* or *j* to go to the next uncovered block, *b*, *p* or *k* for the previous block.

Filter:

File		Statements		Branches		Functions		Lines	
App.jsx		100%	5/5	100%	0/0	100%	3/3	100%	5/5
main.jsx		100%	2/2	100%	0/0	100%	0/0	100%	2/2

```
 ✓ src/__tests__/App.test.jsx (2 tests) 210ms
   ✓ App routing and utilities (2)
     ✓ renders navbar, footer, and matched route 140ms
     ✓ scrolls to top when the location changes 64ms

 Test Files  1 passed (1)
      Tests  2 passed (2)
   Start at  23:22:01
   Duration  7.01s (transform 1.86s, setup 548ms, collect 3.07s, tests 210ms, environment 2.25s,
prepare 64ms)

 PASS  Waiting for file changes...
```

```
nd

 ✓ src/__tests__/main.test.jsx (1 test) 346ms
   ✓ main entrypoint (1)
     ✓ creates a React root and renders the app shell  334ms

 Test Files  1 passed (1)
      Tests  1 passed (1)
   Start at  23:26:13
   Duration  4.60s (transform 441ms, setup 648ms, collect 234ms, tests 346ms, environment 2.50s,
prepare 77ms)

 PASS  Waiting for file changes...
```

Contexts:

- **UserContext:**

```
✓ src/contexts/__tests__/UserContext.test.jsx (9 tests) 374ms
  ✓ UserContext (9)
    ✓ loads current user and exposes it via context 113ms
    ✓ handles missing user response gracefully 18ms
    ✓ allows consumers to update the user via setUser 132ms
    ✓ gracefully handles rejections from getCurrentUser and still resolves loading state
24ms
    ✓ logs an error when getCurrentUser throws synchronously 32ms
    ✓ supports API responses without nested user property 15ms
    ✓ returns null context when useUser is invoked outside the provider 4ms
    ✓ shares updated user state across multiple consumers 24ms
    ✓ skips user updates if the provider unmounts before getCurrentUser resolves 5ms

Test Files  1 passed (1)
     Tests  9 passed (9)
  Start at  02:35:20
  Duration  5.18s (transform 186ms, setup 363ms, collect 871ms, tests 374ms, environment
1.57s, prepare 105ms)
```

All files **src/contexts**

100% Statements 20/20 **100%** Branches 7/7 **100%** Functions 6/6 **100%** Lines 16/16

Press *n* or *j* to go to the next uncovered block, *b*, *p* or *k* for the previous block.

Filter:

File		Statements		Branches		Functions		Lines	
UserContext.jsx		100%	20/20	100%	7/7	100%	6/6	100%	16/16